Interview Questions Based on JD

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What is DevOps

DevOps is a combination of development and operations where both teams work together to build, test, deploy, and monitor applications to faster and reliable

“DevOps is a combination of practices, tools, and processes that automate software development, testing, deployment, and monitoring while improving collaboration between development and operations teams.

What is CI/CD
“CI/CD is a process that automates software build, testing, and deployment. CI focuses on continuously integrating and testing code changes, while CD automates delivering or deploying the tested application to the target environment.”

Stages

1. Source / Code
Developer writes code and pushes it to **GitHub/GitLab/Azure Repos**.

Developer → Git → GitHub
Tools: Git, GitHub, GitLab, Azure Repos

2. Build
The pipeline takes the source code and **builds the application**.
For example:

- Java → Maven / Gradle
- Python → pip / build tools
- .NET → dotnet build

Source Code → Build → Application Package

Build
What: The code is compiled and turned into something runnable — a binary, a Docker image, a JAR, etc.
Why: Raw source code isn't runnable; it needs to be assembled.
Tool examples: Maven, Gradle, npm, Docker

3 Test

What: Automated tests run to check the code works.

Usually in layers:

- Unit tests – Does each small function work?
- Integration tests – Do the pieces work together?
- End-to-end tests – Does the whole app work like a user would use it?

Why: Catch bugs before they reach real users.

Tool examples: Jest, PyTest, JUnit, Selenium, Cypress

🍕 Analogy: The chef tastes the pizza before it leaves the kitchen.

4 Package / Artifact

What: The tested, working code is bundled into a versioned artifact (Docker image, zip, etc.) and stored.

Why: So the exact same tested thing gets deployed everywhere (no "works on my machine" surprises).

Tool examples: Docker Registry, Nexus, Artifactory, GitHub Packages

🍕 Analogy: The pizza goes into a sealed box with a label.

5 Deploy
The application is deployed to an environment.

Docker Image
↓
AWS / Azure
↓
Application Running

Typical environments:
Development → Testing → Staging → Production
Deployment can be automated using **Jenkins, Azure DevOps, GitHub Actions, Kubernetes**, etc.

 Analogy: A test customer tries the pizza before it goes to the real customer.

Deploy to Staging

What: The artifact is deployed to a test environment that mimics production.

Why: Last chance to catch issues before real users see them.

Tool examples: Kubernetes, AWS, Terraform, Helm

🍕 Analogy: A test customer tries the pizza before it goes to the real customer.

6 Approval (optional)

What: A human clicks "Approve" to move to production. (This is what separates Continuous Delivery from Continuous Deployment.)

Why: Some teams want a final safety check before going live.

Tool examples: GitHub Environments, Jenkins approval gates

🍕 Analogy: The manager inspects the box before it's delivered.

7️⃣ Deploy to Production

What: The app goes live for real users.

Why: This is the whole point — new features reach customers.

Techniques:

- Blue-Green – Two identical environments, switch traffic
- Canary – Send 5% of users first, then roll out
- Rolling – Update servers one at a time

🍕 Analogy: The pizza is delivered to the customer's door.

8️⃣ Monitor & Feedback

What: Watch logs, metrics, errors, and performance after deploy.

Why: If something breaks, you roll back fast and learn why.

Tool examples: Prometheus, Grafana, Datadog, Sentry

🍕 Analogy: You call the customer — "How was the pizza?" 🍕

What is Azure DevOps
Azure DevOps is a set of development and DevOps services provided by Microsoft. It includes the Azure Repos to source control and Azure Pipelines to CI/CD, Azure bords to work tracking and Azure test plans to testing.

What is a build agent?
An agent is the machine that executes pipeline jobs. It runs tasks such as downloading source code, installing dependencies, building applications, running tests, and deploying resources.

What is YAML in Azure DevOps?
YAML is a configuration file used to define a pipeline as code. It describes triggers, stages, jobs, steps, variables, and tasks.

What is GitHub Actions?
GitHub Actions is a CI/CD automation service integrated with GitHub. It can automatically build, test, and deploy applications when events such as code pushes or pull requests occur.

Azure DevOps vs GitHub Actions?
A good junior-level answer:
Both can implement CI/CD. Azure DevOps provides a broader DevOps platform including Boards, Repos, Pipelines, Artifacts, and Test Plans. GitHub Actions is tightly integrated with GitHub repositories and uses workflow YAML files for automation.

What is Terraform?
Terraform is an Infrastructure as Code tool developed by HashiCorp. It allows us to define and provision infrastructure using configuration files instead of manually creating resources through the cloud portal.

Why do we use Terraform?
Mention:

- Automation
- Consistency
- Repeatability
- Version control
- Infrastructure standardization
- Easy resource provisioning

What is Infrastructure as Code?
Infrastructure as Code means managing infrastructure through configuration files and code rather than manually creating resources.

What are the basic Terraform commands?
Memorize these:

terraform init
terraform validate
terraform plan
terraform apply
terraform destroy
Explain:

- terraform init → initializes the working directory
- terraform validate → validates configuration
- terraform plan → shows what Terraform will create/change/delete
- terraform apply → applies the configuration
- terraform destroy → removes resources managed by Terraform

What is Terraform state?

Terraform state is a file that keeps track of the infrastructure resources Terraform manages and their current known state.

terraform.tfstate

What is Terraform provider?

A provider allows Terraform to communicate with a particular platform or service.
For Azure:

```
provider "azurerm" {
 features {}
}
```

You can say:

Azure Resource Manager is accessed through the AzureRM Terraform provider.

What is Terraform plan?

terraform plan compares the desired infrastructure configuration with the current state and shows the changes Terraform intends to make.
Very important: plan does not normally create the resources.

5. Bicep and ARM Templates

Q18. What is Bicep?

Bicep is Microsoft's Infrastructure as Code language for deploying Azure resources. It provides a simpler syntax compared with ARM JSON templates.

Q19. What is an ARM template?

An ARM template is a JSON-based Infrastructure as Code template used to define and deploy

Azure resources.

Q20. Terraform vs Bicep?

A good answer:

Terraform is a multi-cloud Infrastructure as Code tool and can work with Azure, AWS, and other platforms. Bicep is Microsoft's domain-specific language designed specifically for Azure resource deployment.

Q21. Bicep vs ARM?

Bicep is essentially a simpler authoring language for Azure infrastructure. ARM templates use JSON and can become difficult to maintain for large deployments, while Bicep provides cleaner syntax and is transpiled into ARM-compatible deployment definitions.

6. Deployment troubleshooting

This section is **very important for a junior DevOps role**.

Q22. Deployment failed. What will you do?

Don't say:

"I will restart the server."

Instead give a structured answer:

First, I would check the pipeline logs to identify the exact error. Then I would determine whether the problem is related to source code, build dependencies, authentication, infrastructure, networking, or application configuration. I would reproduce the issue if possible, make the required correction, test it in a lower environment, and then redeploy.

What if Azure DevOps pipeline cannot connect to Azure?

Possible causes:

- Service connection issue
- Expired credentials
- Incorrect permissions
- Wrong subscription
- RBAC problem
- Network restrictions

Answer:

I would first check the pipeline logs and Azure service connection. Then I would verify authentication, subscription, service principal permissions, RBAC roles, and any network restrictions.